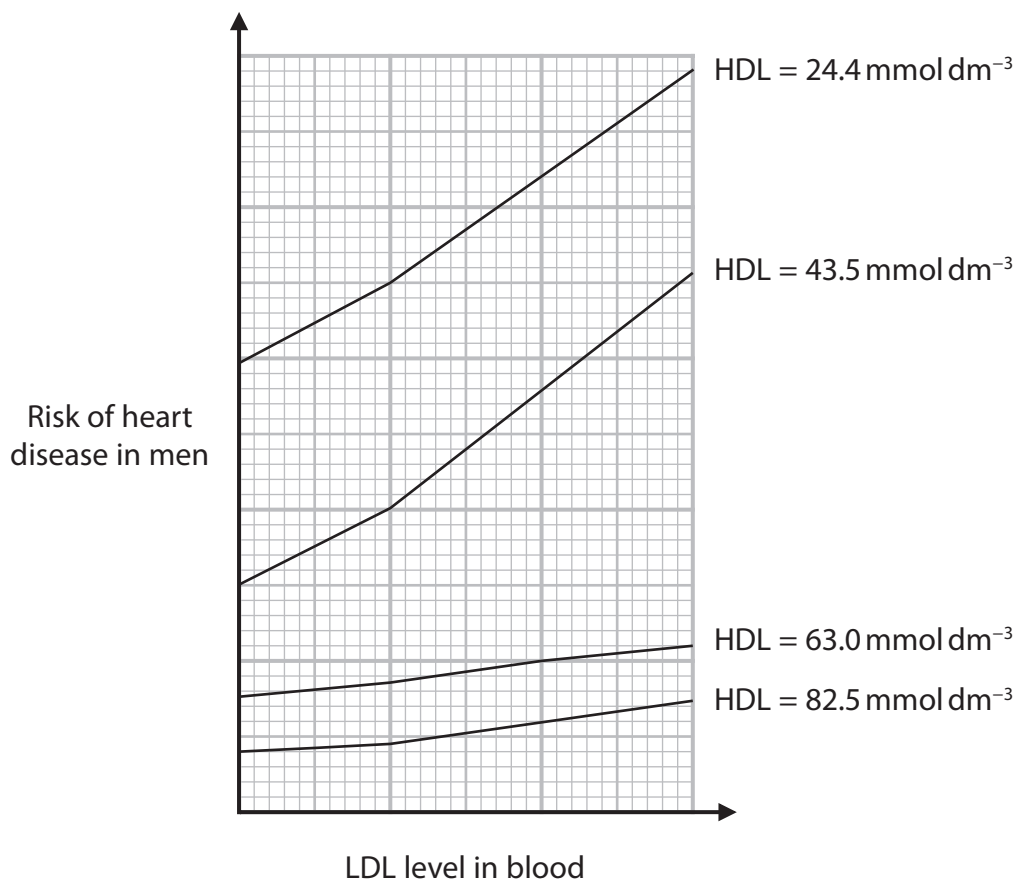


7 A number of factors affect the risk of a person developing heart disease.

One factor affecting this risk is the level of high-density lipoprotein (HDL) in the blood.

*(a) The graph shows the effects of HDL levels and low-density lipoprotein (LDL) levels in the blood on the risk of men developing heart disease.



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For men, a blood HDL level greater than 40 mg dm^{-3} is thought to be optimal.

Explain why a man with a blood HDL level greater than 40 mg dm^{-3} may still have a high risk of developing heart disease.

Use the information in the graph and your own knowledge to support your answer.

(6)

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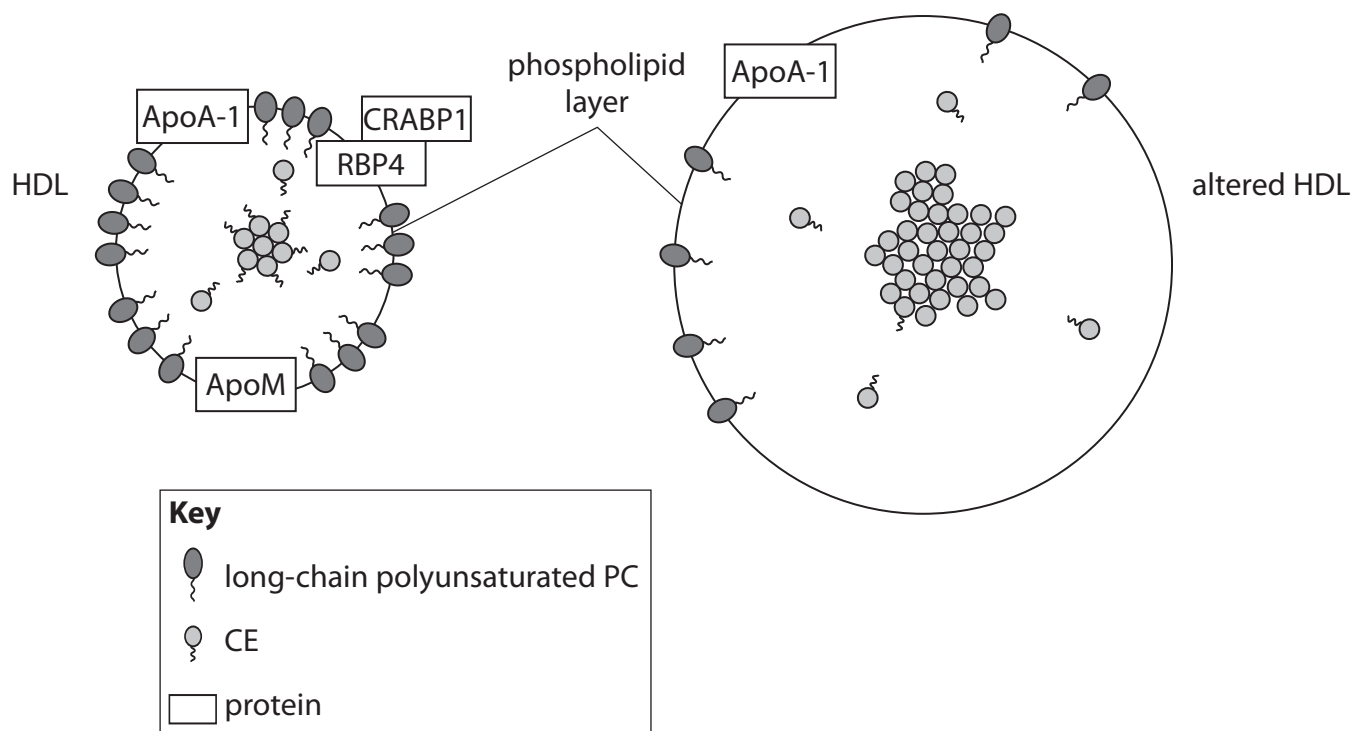
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- (b) Very high levels of cholesterol in the blood can alter the structure of HDL. This altered HDL is less effective in reducing the risk of heart disease.

The diagram shows the structure of HDL in blood with a low level of cholesterol and altered HDL in blood with a high level of cholesterol.



(Source: <https://www.sciencedirect.com/science/article/pii/S0735109717373448>)

- (i) Compare and contrast the structure of HDL with altered HDL.

(3)

(ii) The antioxidant properties of altered HDL are reduced.

Explain the effect that this has on reducing the risk of heart disease.

(3)

(Total for Question 7 = 12 marks)

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